

Summary Sheet

Executable evidence-gate specification for pharmacogenomic LLM-alert review

Author: Ravi Bajaj | Course: AI in Healthcare

Conference / Symposium

Field	Value
Name	Machine Learning for Health Symposium (ML4H 2026)
URL	https://ml4h.ahli.cc/
Author instructions / CFP	https://ml4h.ahli.cc/submit/call-for-papers/
Submission deadline	September 10, 2026, 11:59 PM AoE
Author notification	October 22, 2026
Camera-ready deadline	November 7, 2026 (tentative)
Event dates	December 6-7, 2026
Location	Sydney, Australia
Suggested track	Findings Track for preliminary specification-conformance work; Proceedings Track only after independent review.
Editable paper format	LaTeX/Overleaf source: paper/ml4h_findings_evidence_gate.tex with paper/ml4h_findings_refs.bib; compiled PDF: paper/ml4h_findings_evidence_gate.pdf.

Title and Abstract

Title: An Executable Evidence-Gate Specification for Pharmacogenomic LLM Alerts

Abstract: LLM-generated medication alerts can sound more certain than their evidence permits. We built an executable Stage 1 specification for pharmacogenomic and genomic medication-alert text. Given structured annotations that a future live system would need to extract, the specification checks source support, population fit, and claim strength, then returns allow, narrow, abstain, or deny. Stage 1 uses author-designed synthetic cases with oracle annotations; it is a conformance and preregistration scaffold, not an end-to-end clinical system. Disabling source support, population fit, or claim strength let 2, 1, and 6 of 23 designed overclaims pass unchanged, respectively. A claim-strength-only comparator left 3/23 overclaims unchanged (PGX31-PGX33), while the full monitor routed 23/23 and preserved 10/10 bounded alerts. Arm B withheld source-support labels for PGX31-PGX33; GPT-5.6-terra matched all three deterministic actions and primary checks, while Grok-4.5 matched PGX31 and PGX32 but attributed PGX33 to claim strength. These are specification-conformance results, not clinical safety, annotation validity, generalization, or patient-benefit evidence.

Generative AI Use

Generative AI tools were used for brainstorming, critique, code scaffolding, drafting, packaging, formatting, and LLM self-evaluation baselines on synthetic annotations only. Arm A used GPT-5.6-terra and Grok-4.5 with full structured labels; Arm B used the same models on PGX31-PGX33 with source-support labels withheld, sentinel cues removed, and structured citations supplied. No real patient records, protected health information, or private genomic data were transmitted. The author selected the final scope, directed the pruning, verified sources, reviewed generated text, ran local tests, and remains responsible for all claims, limitations, and submission materials.

Optional Data, Code, and Tools

- Synthetic 33-case pharmacogenomic/genomic medication-alert case bank with no real patient data, PHI, diagnosis, or treatment recommendation.
- Executable three-check controller with unit tests and deterministic replay outputs.
- One-check ablation CSV, simple policy-comparator CSV, precedence-sensitivity CSV, Arm A/Arm B LLM self-evaluation CSV/JSON outputs, synthetic uniform-NARROW sanity outputs, and CSV/JSON summaries for overclaim handling, inappropriate denial, calibration, and check categories.

Evidence-Boundary Framework

The empirical workflow is pharmacogenomics, but the design pattern is broader: source support asks whether evidence exists and applies; population fit asks whether evidence transports to the target patient or population; and claim strength asks whether the evidence supports the displayed action rather than only a weaker association or surrogate endpoint.